



Image Convolution

Introducción a métodos de Machine Learning para ciencias
2023

What is a digital image?

$$f: \mathbb{N}^d \rightarrow \mathbb{R}^B$$
$$m, n, \dots \mapsto f(m, n, \dots) = \{i_1, \dots, i_B\}.$$

- For example, a **grayscale image** corresponds to $d = 2$ (the image has two dimensions) and $B = 1$ (there is only one value per location (m, n) : the grayscale intensity).
- A common color image corresponds to $d = 2$ and $B = 3$ bands, typically red, green, and blue.
- An **MRI image** corresponds to $d = 3$ (the image is three-dimensional) and $B = 1$.

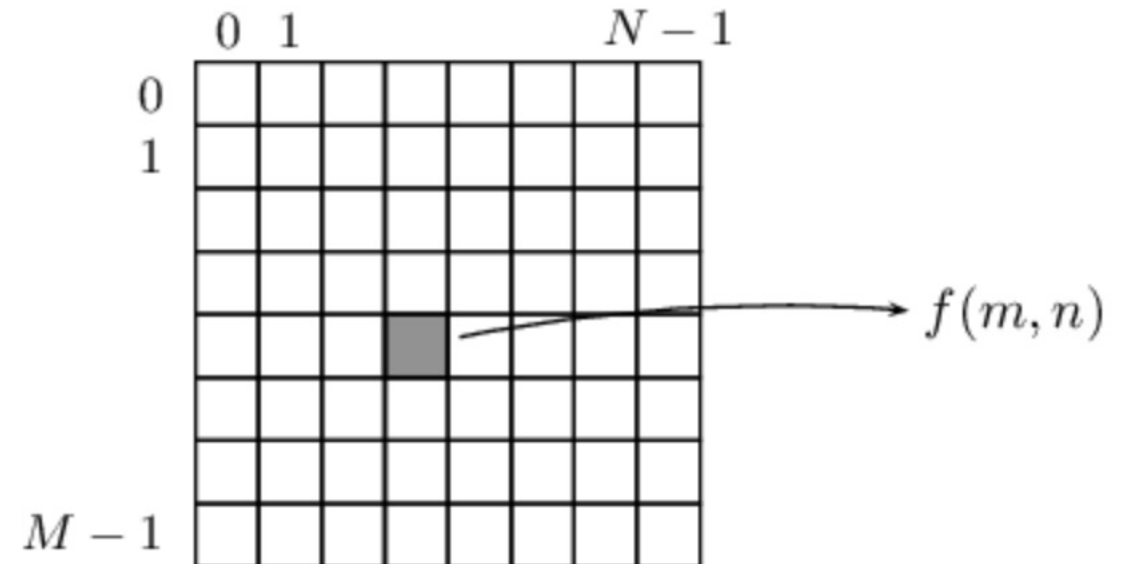
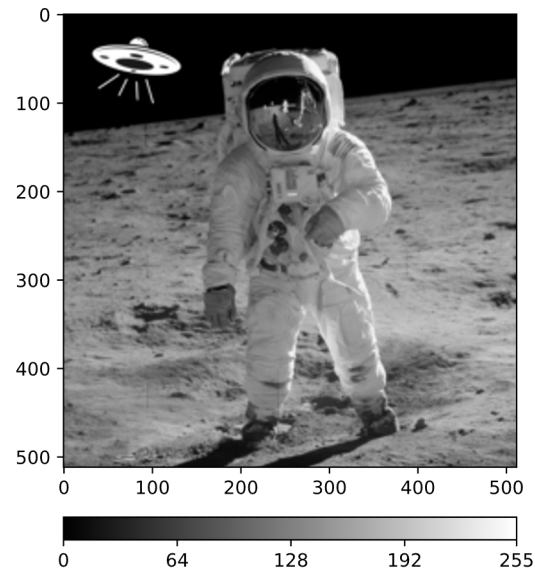
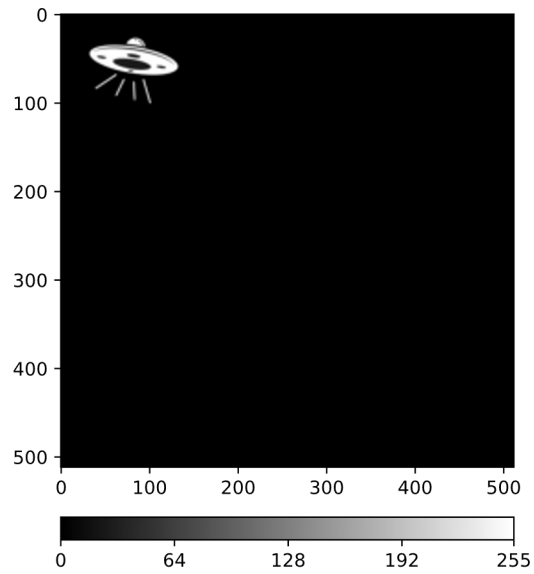
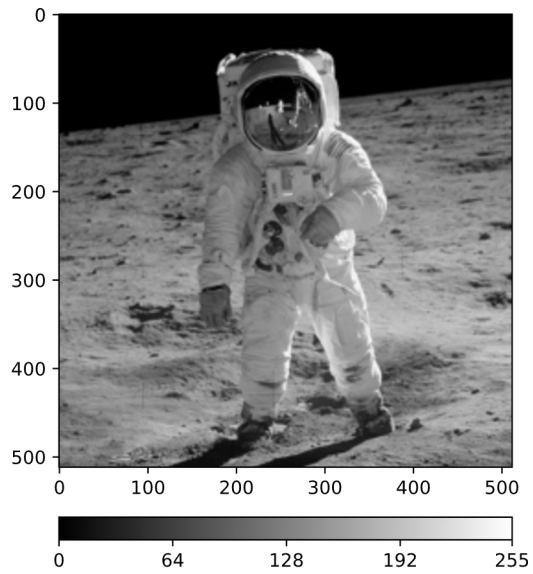


Fig. 1 Coordinate system generally used in image processing.

Arithmetic operations of images

Addition

$$\forall m, n, \quad h(m, n) = f(m, n) + g(m, n).$$



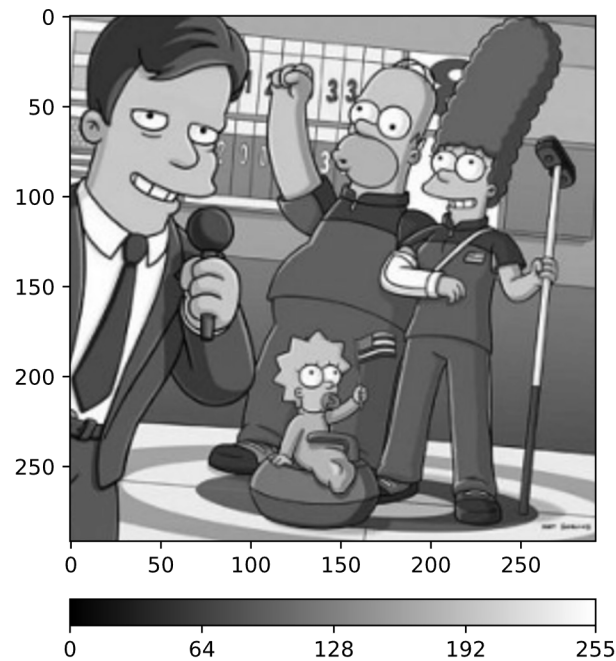
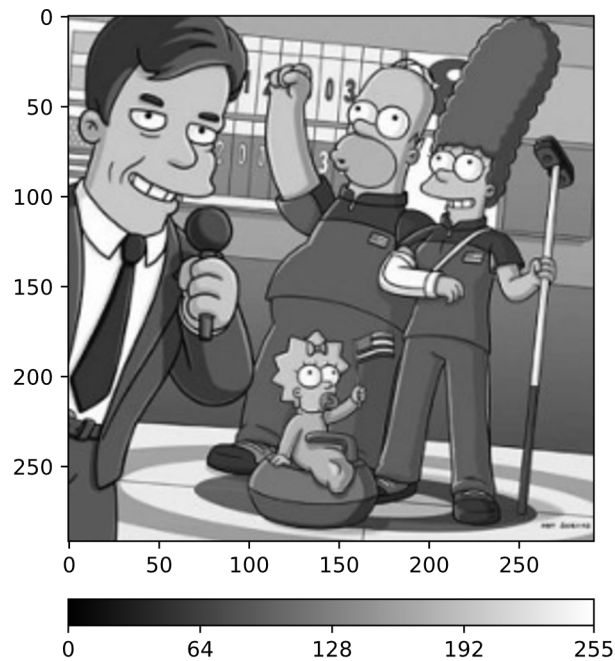
Arithmetic operations of images

subtraction

or

$$\forall m, n, \quad h(m, n) = f(m, n) - g(m, n)$$

$$\forall m, n, \quad h(m, n) = |f(m, n) - g(m, n)|$$



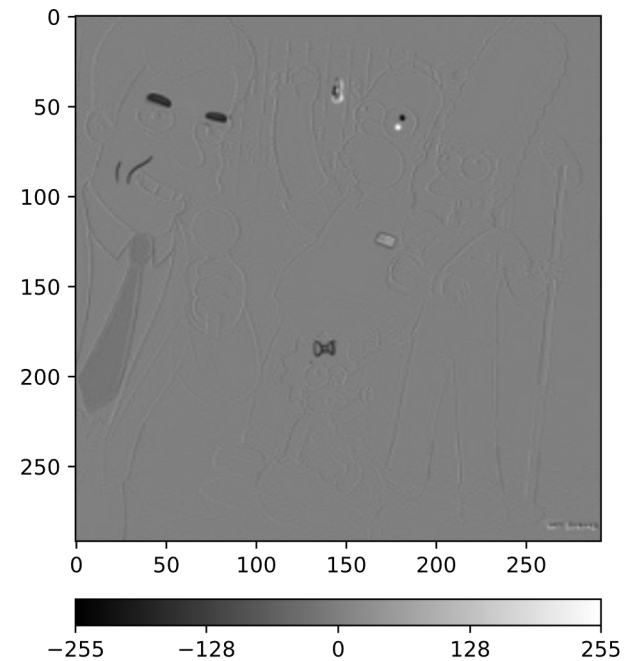
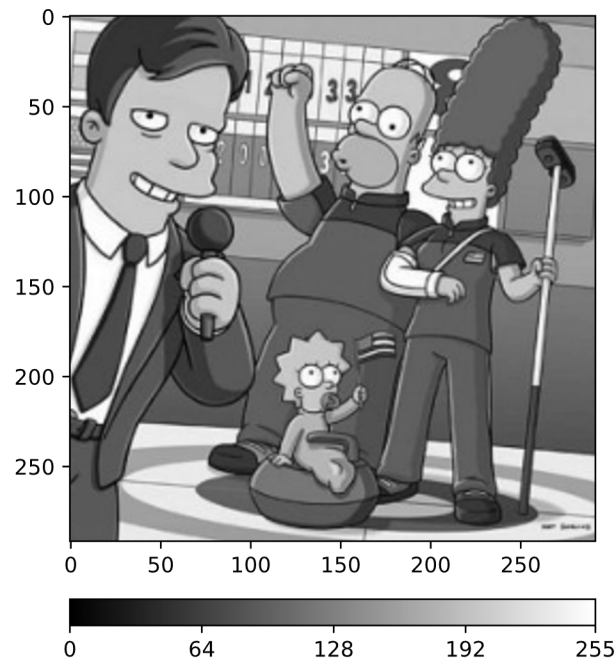
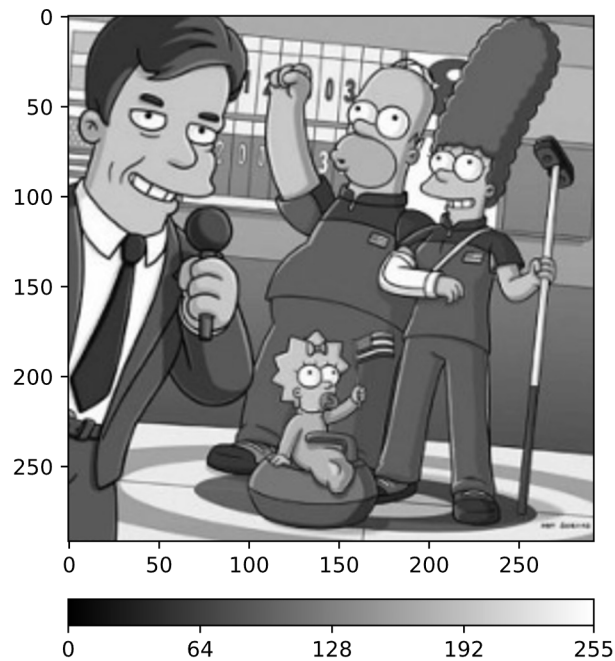
Arithmetic operations of images

subtraction

or

$$\forall m, n, \quad h(m, n) = f(m, n) - g(m, n)$$

$$\forall m, n, \quad h(m, n) = |f(m, n) - g(m, n)|$$



Arithmetic operations of images

Division

$$\forall m, n, \quad h(m, n) = \frac{f(m, n)}{g(m, n)}.$$

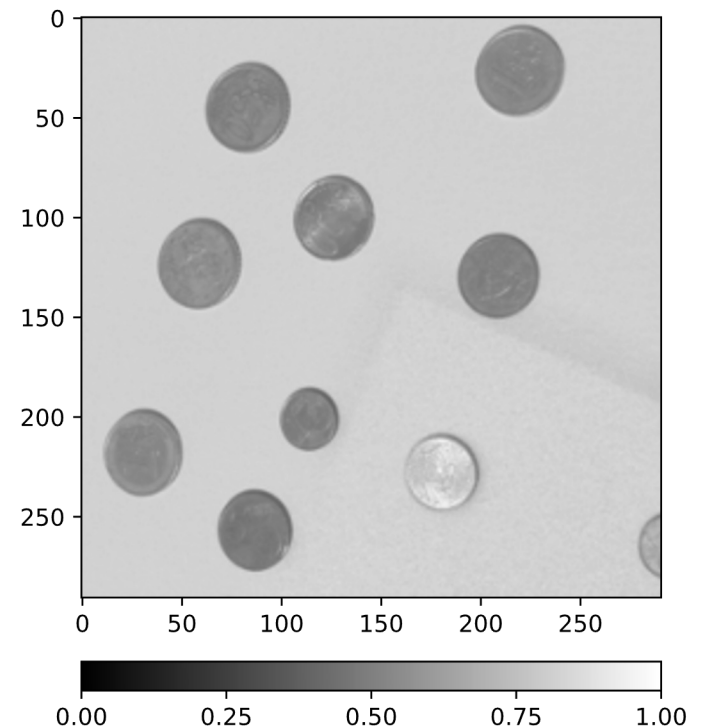
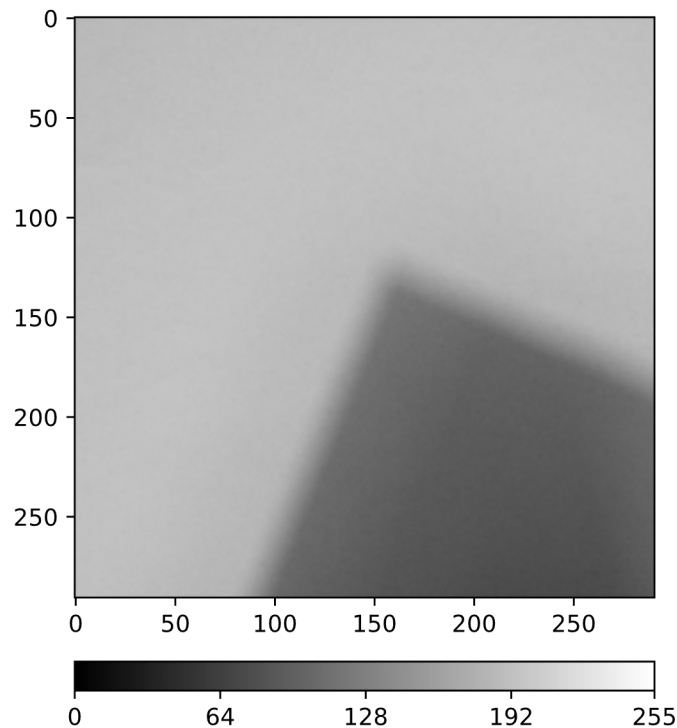
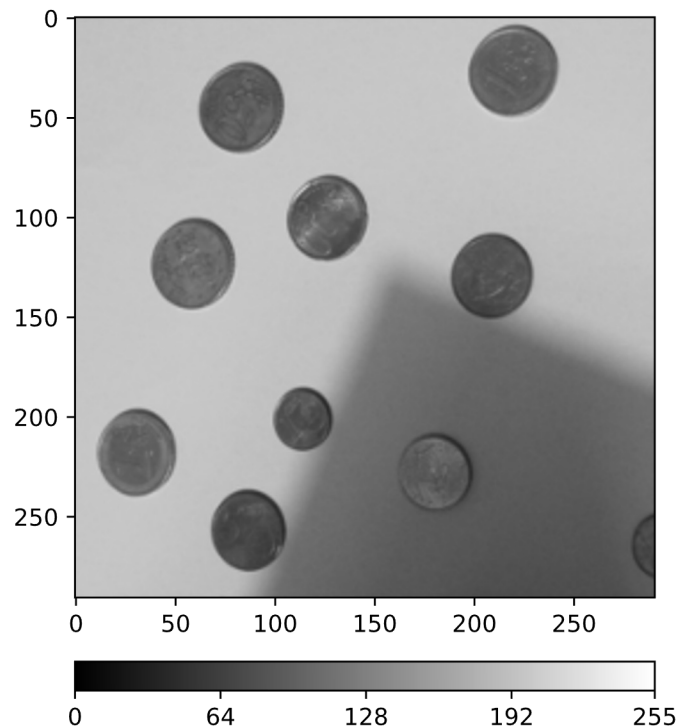
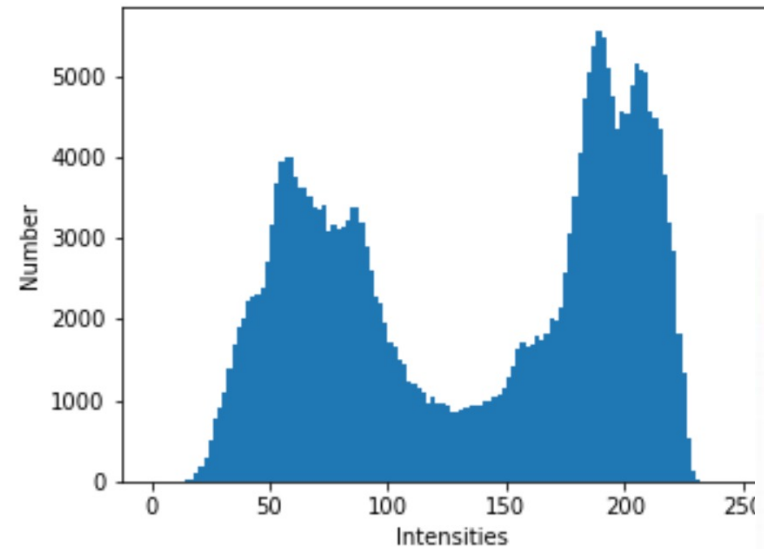


Image histogram



Histogram Transformation

An histogram transformation consists in applying a mathematical function to the intensity distribution.

Generally, the transformations are useful to improve the visual quality of an image, but are rarely needed inside an automatic processing.

$$j = T(i)$$

we suppose the pixel intensities to lie in $[0, 1]$

Some common transformations:

- Negative image
- Gamma correction
- Histogram spreading
- Histogram equalization



Negative image

we suppose the pixel intensities to lie in $[0, 1]$

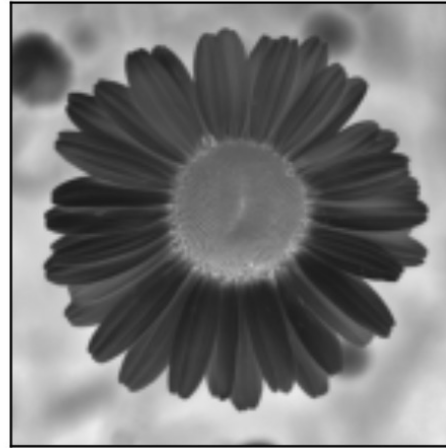
$$T(i) = 1 - i$$

Negative image: the gray levels are reversed

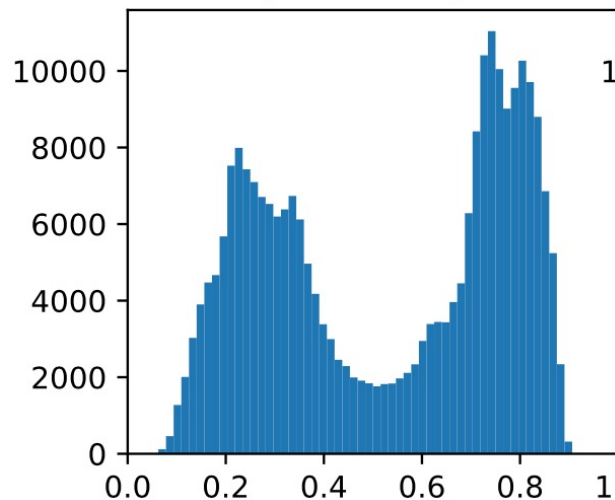
Original image



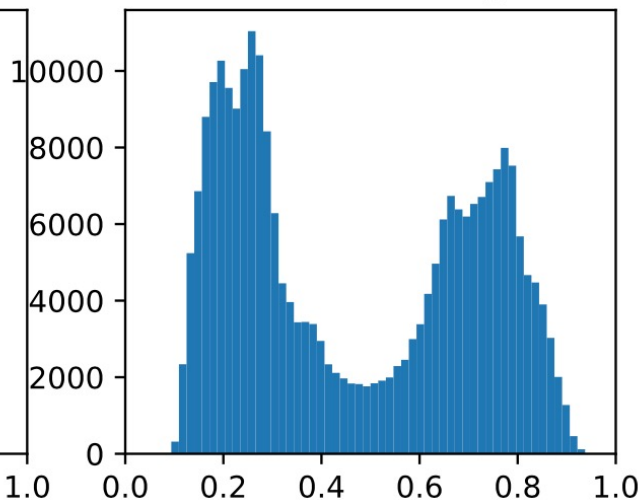
Transformed image



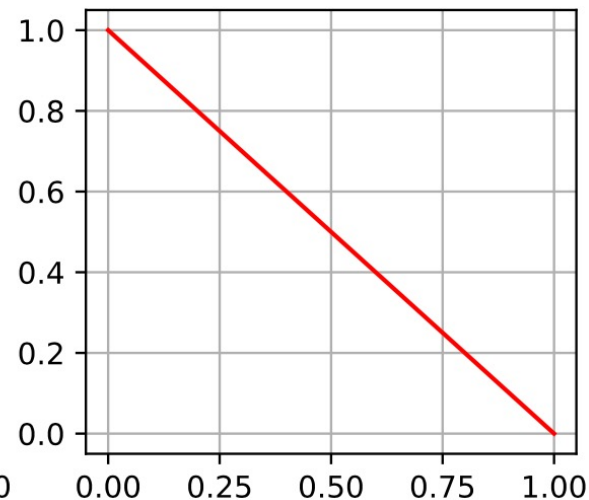
Original histogram



Transformed histogram



Transformation



Gamma correction

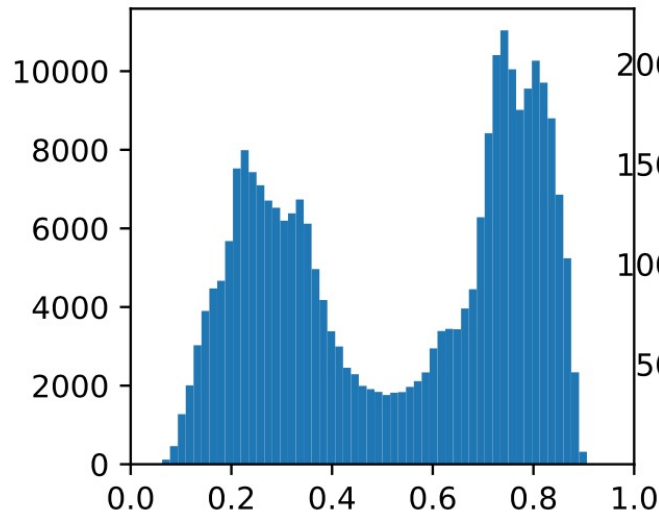
Original image



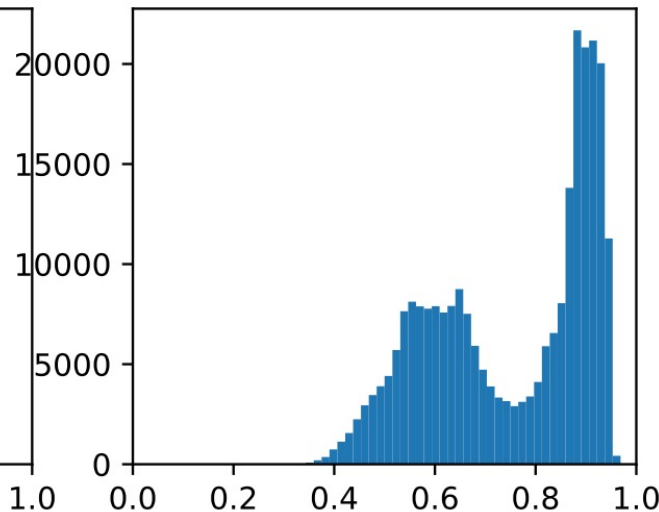
Transformed image



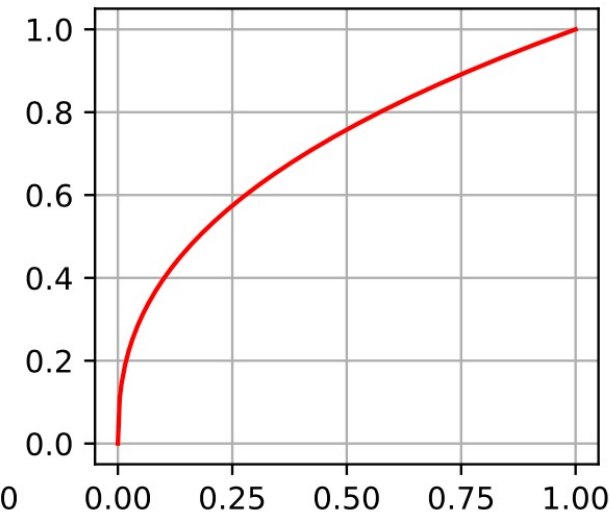
Original histogram



Transformed histogram



Transformation



we suppose the pixel intensities to lie in $[0, 1]$

$$T(i) = i^\gamma$$

Gamma correction modifies the colors of an image acquired by an electronic system.

It is used to consider the non-linear sensibility of human eyes to the light. Here, $\gamma=0.4$

Histogram spreading

we suppose the pixel intensities to lie in $[0, 1]$

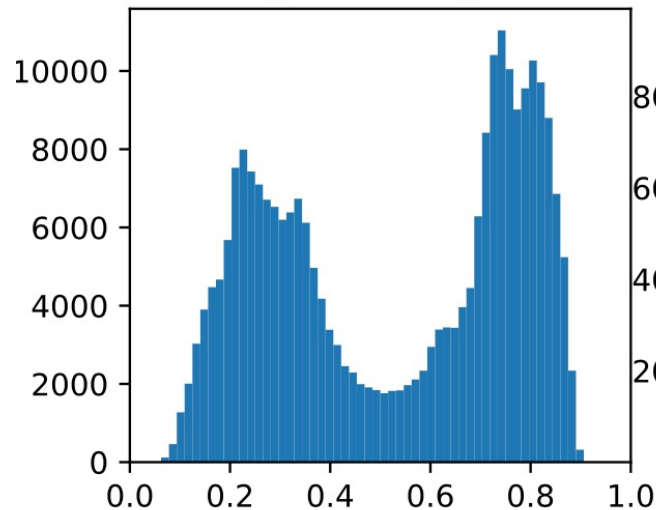
Original image



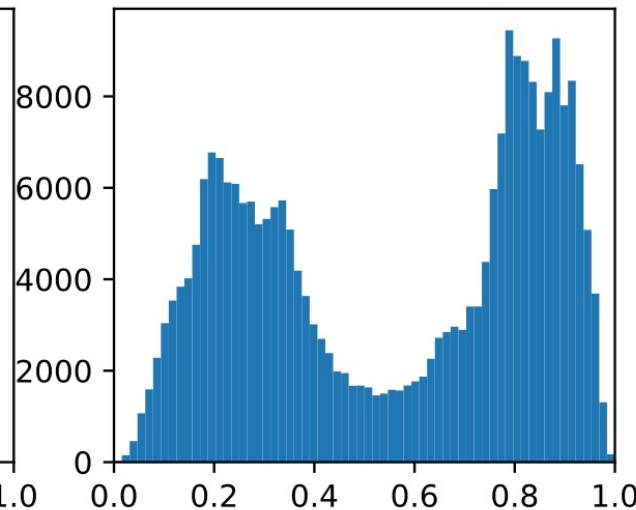
Transformed image



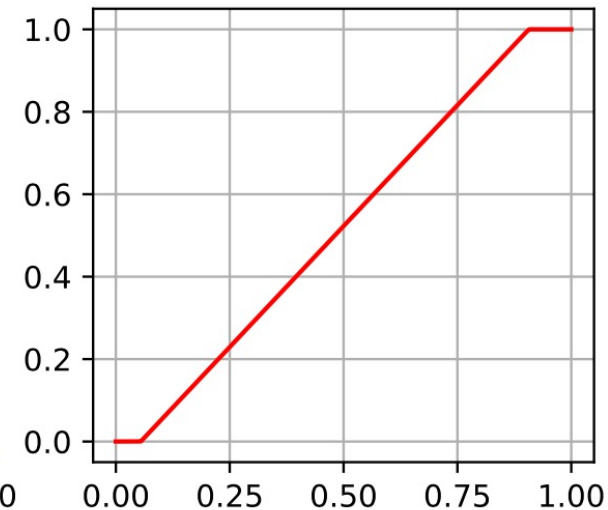
Original histogram



Transformed histogram



Transformation



$$T(i) = \frac{i - i_{\min}}{i_{\max} - i_{\min}}$$

Histogram spreading enhances the contrast by “dilating” the histogram to the whole intensity interval.

Equalization Histogram

we suppose the pixel intensities to lie in $[0, 1]$

$$T(i) = \frac{1}{MN} \sum_{k=0}^i n_k$$

- where M and N are the image size and n_k is the number of pixels with intensity i_k .
- This transformation aims to spread the histogram over the entire intensity range, and to make the histogram as flat as possible.
- A consequence is an increasing of the image contrast

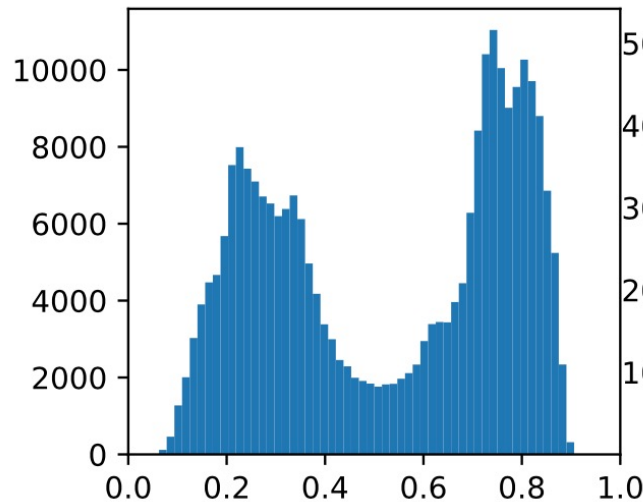
Original image



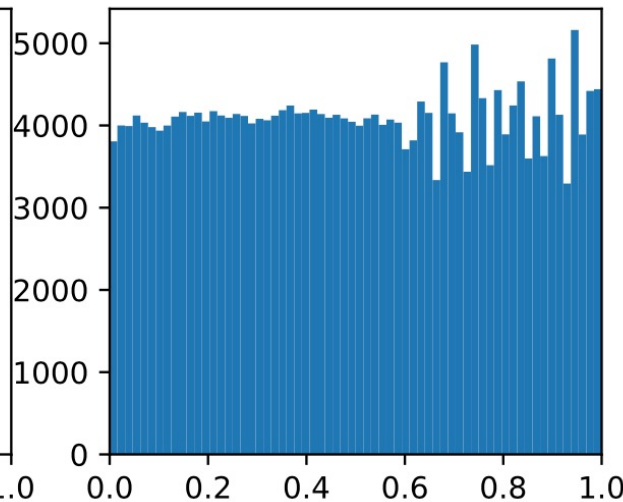
Transformed image



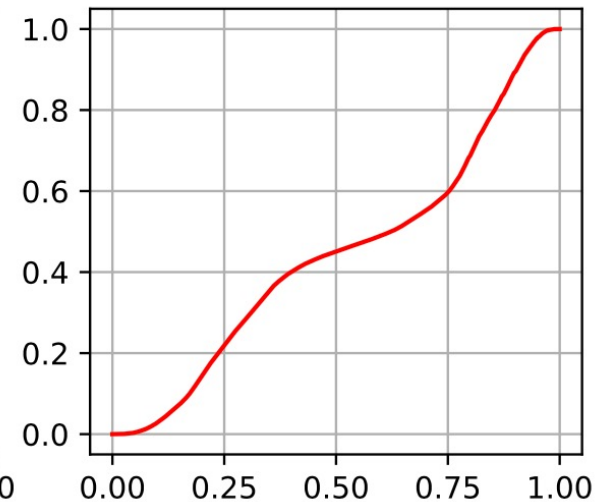
Original histogram



Transformed histogram



Transformation

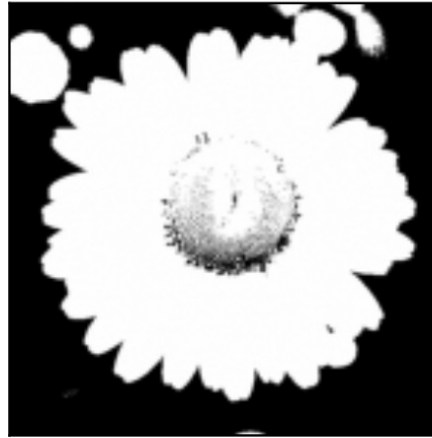


Thresholding

Original image

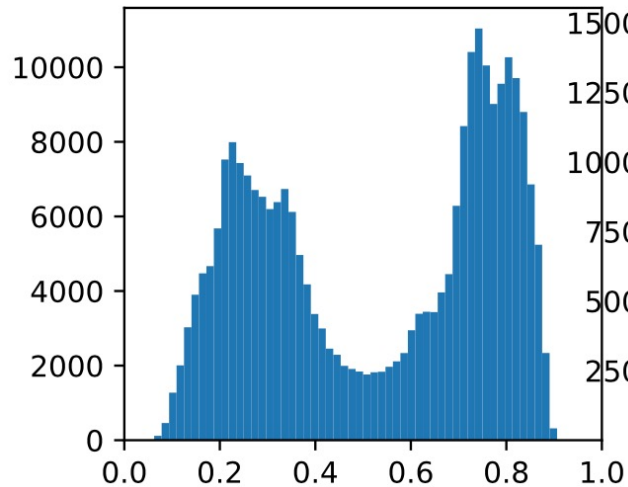


Transformed image

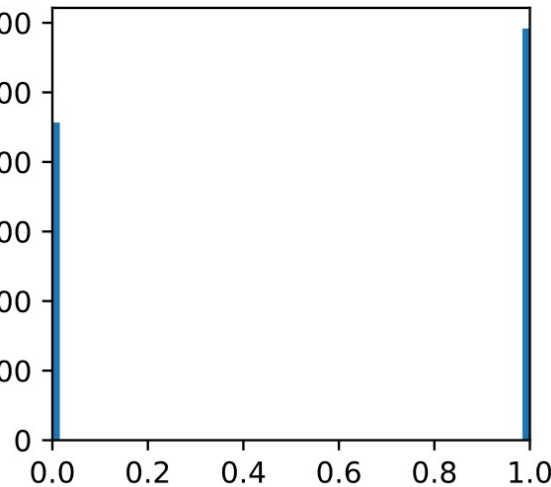


Threshold with a threshold set to 0.45

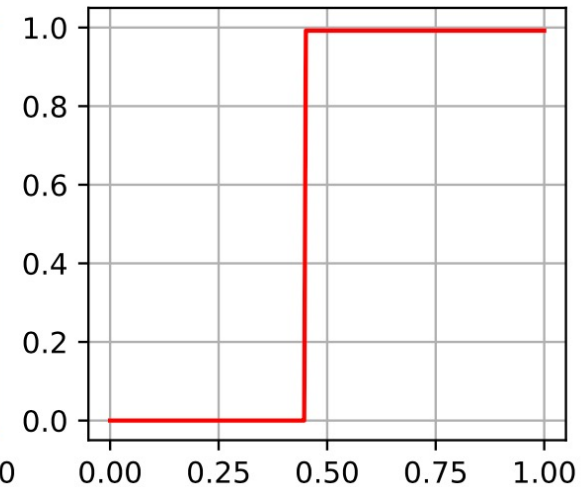
Original histogram



Transformed histogram



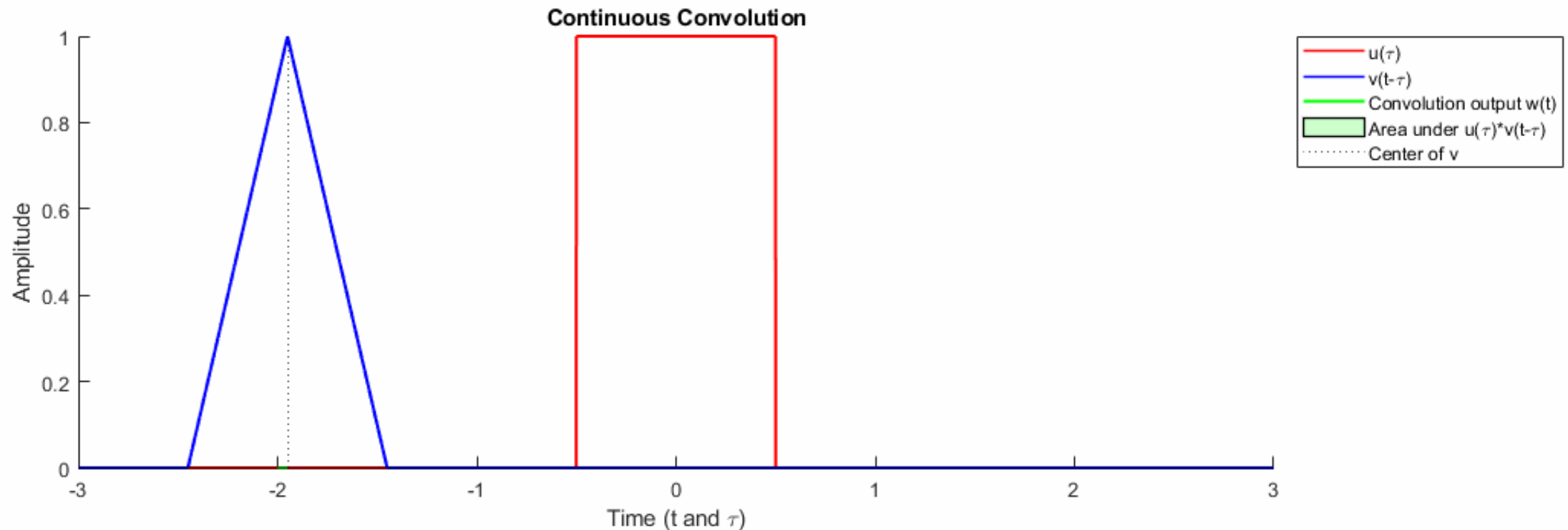
Transformation

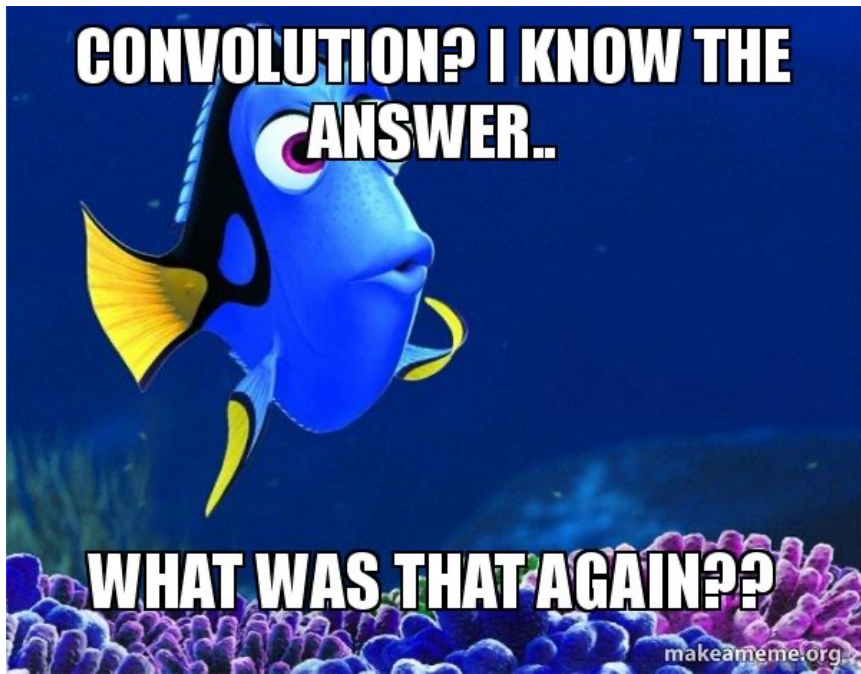


Convolution

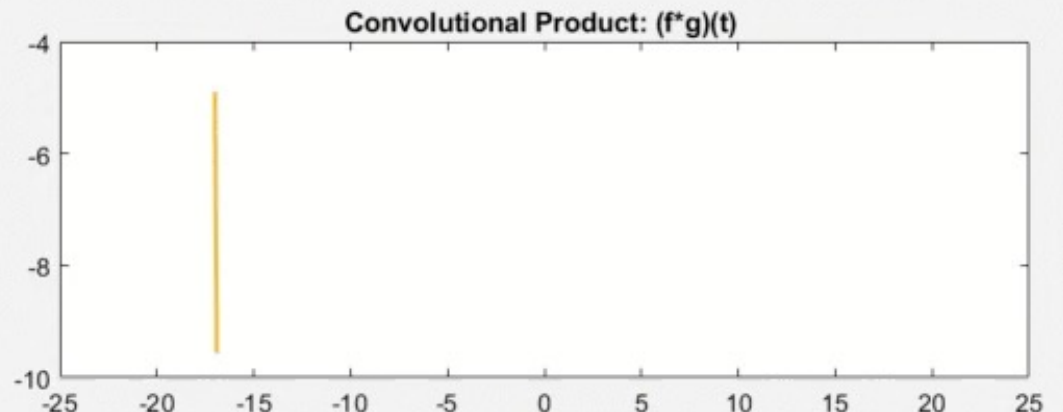
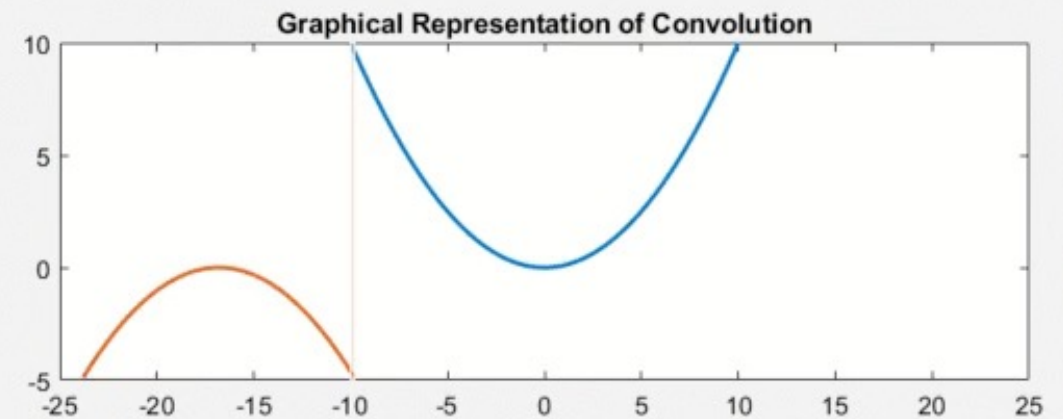
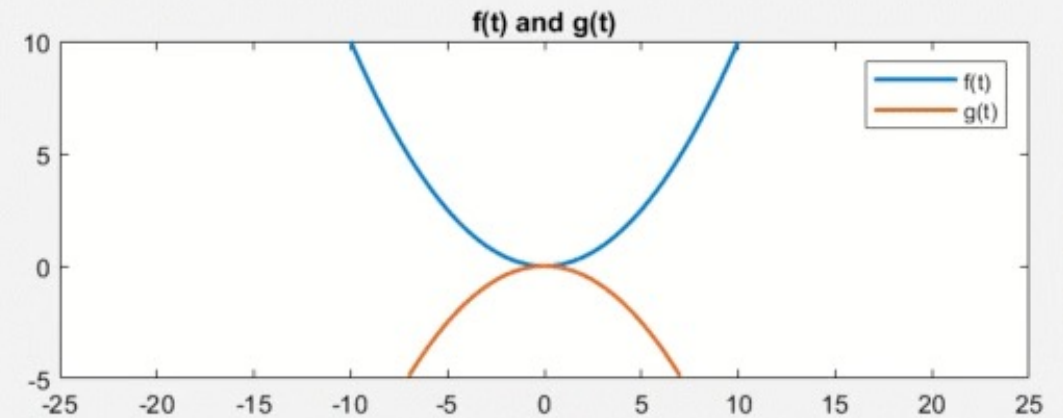
Formally, convolution is an integral that expresses the amount of overlap of one function, $f(t)$, as it is shifted over function $g(t)$, expressed as:

$$(f * g)(t) \approx^{def} \int_{-\infty}^{\infty} f(\tau)g(t - \tau)d\tau$$





- Convolution is a mathematical operation that combines two functions to describe the overlap between them.
- Convolution takes two functions and “slides” one of them over the other, multiplying the function values at each point where they overlap, and adding up the products to create a new function.
- This process creates a new function that represents how the two original functions interact with each other

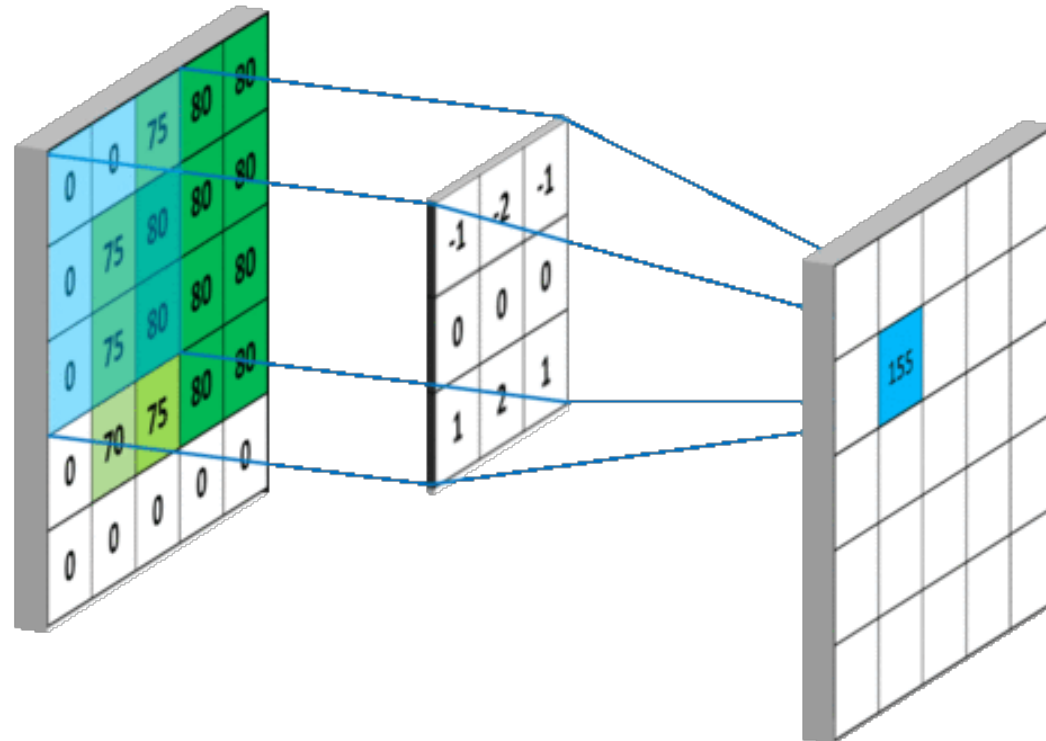


Convolution in Image Processing

In image processing, convolutional filtering can be used to implement algorithms such as:

1. edge detection,
2. image sharpening, and
3. image blurring.

This is done by selecting the appropriate kernel (convolution matrix).



Sharpening Filter Kernel

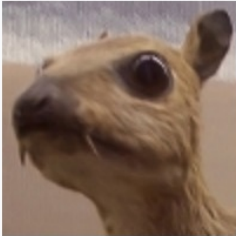
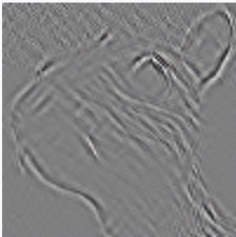
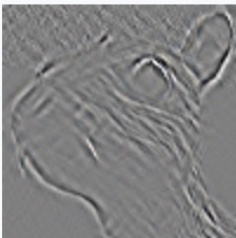
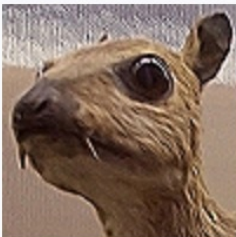
0	-1	0
-1	5	-1
0	-1	0

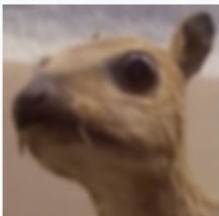
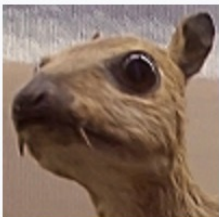
Input

0	0	0	0	0	0	0
0	1	2	2	1	0	0
0	1	2	1	2	1	0
0	0	2	3	2	0	0
0	0	5	2	2	0	0
0	0	7	2	1	1	0
0	0	0	0	0	0	0

Sharpened Output

2	5	6	1	-2
2	4	-4	5	3
-3	0	8	3	-3
-5	14	-2	5	-3
-7	28	0	0	4

Operation	Kernel ω	Image result $g(x,y)$
Identity	$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$	
Ridge or edge detection	$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix}$	
	$\begin{bmatrix} -1 & -1 & -1 \\ -1 & 8 & -1 \\ -1 & -1 & -1 \end{bmatrix}$	
Sharpen	$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 5 & -1 \\ 0 & -1 & 0 \end{bmatrix}$	

Box blur (normalized)	$\frac{1}{9} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$	
Gaussian blur 3 × 3 (approximation)	$\frac{1}{16} \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$	
Gaussian blur 5 × 5 (approximation)	$\frac{1}{256} \begin{bmatrix} 1 & 4 & 6 & 4 & 1 \\ 4 & 16 & 24 & 16 & 4 \\ 6 & 24 & 36 & 24 & 6 \\ 4 & 16 & 24 & 16 & 4 \\ 1 & 4 & 6 & 4 & 1 \end{bmatrix}$	
Unsharp masking 5 × 5 Based on Gaussian blur with amount as 1 and threshold as 0 (with no image mask)	$\frac{-1}{256} \begin{bmatrix} 1 & 4 & 6 & 4 & 1 \\ 4 & 16 & 24 & 16 & 4 \\ 6 & 24 & -476 & 24 & 6 \\ 4 & 16 & 24 & 16 & 4 \\ 1 & 4 & 6 & 4 & 1 \end{bmatrix}$	

```
for each image row in input image:  
  for each pixel in image row:  
  
    set accumulator to zero  
  
    for each kernel row in kernel:  
      for each element in kernel row:  
  
        if element position corresponding* to pixel position then  
          multiply element value corresponding* to pixel value  
          add result to accumulator  
        endif  
  
    set output image pixel to accumulator
```



Tarea:

Parte 1:

- A una imagen, aplicar todos los métodos de transformación de histogramas vistos en clase.

Parte 2:

- Implemente el algoritmo de convolución de una imagen mostrado en clase.

